Opt-in Kummer signed-driver source review

All pinned source-bindings match retained originals. Native prefix containing advance and signed_fallback is byte-identical; signed lift, walk, graph and Labels files remain unchanged. Existing suffix fixtures also match exactly.

Literal run loop retains RNG draws, arithmetic calls, label updates, gate/restart order, first signed-divisor deduplication and original graph labels. New canonical temporary evaluates the same original d.min(j.neg(d)) once. No Labels::ingest insertion or graph algorithm replacement occurs.

Disabled observation skips JSON event construction and observed call increment; this is a source property, not zero-overhead or machine-code equivalence evidence. Wrapper/function calls and observed serialization costs require measurement.

Both actual advance sites and signed lift receive begin/result pairs. End callbacks occur before following counter/label mutation; callback errors propagate through ? and stop the invocation with incomplete Trace prefix. Relation callback precedes graph.edge; graph-result callback follows it. Terminal callback precedes original result-row serialization.

NativeEvents driver state checks call numbers, matching operation, original lift labels/divisor, required relation/duplicate disposition and graph-result completion; terminal attempts/calls/recovered must match. Trace core retains original limits, flush and first-failure poisoning. This validates diagnostic structure, not geometry or global scalar invariants.

Fixed producer controls stage archived original as pruned-original.rs under test-only module and compare the complete original/disabled/observed output row, including field/walk/recovery/graph counts. Header construction matches fixed fixture initial labels0,1 and pair cases2,3, exact seed/cap and validation-disabled flag.

Eleven planned target invocations are original+disabled+observed+eight callback writer faults. Missing fixed-target duplicate coverage fails rather than adding seeds. Each injected fault requires immediate error, poisoned incomplete Trace and observed callback count position+1.

Separate actual native forced-zero-difference advance verifies fallback reconstruction; zero-vector lift requires None/zero_projective_vector/tested0. Four protocol negatives cover missing result, wrong call number, changed signed labels and omitted lift disposition.

No compile, tests, target cohort or timing comparison was executed. Native target cap1 and serial exact-test invocation are staging requirements; the previous blanket main would otherwise retain1260targets. Runtime plan, staging/lockfile publication and finite admission remain separate.

Coverage and result boundaries

delta_zero, D_zero, degree-one ambiguity and restart projection failure have hooks but no new fixed native demonstrations.

Driver faults use write failures at eight joints, not new driver-specific flush fault cases; inherited Trace flush semantics are unchanged.

Adapter accepts recorded numeric node/sign fields and canonical objects but does not independently recompute canonicalization, graph equations or scalar labels.

Terminal diagnostic acceptance precedes file write; a complete Trace alone does not establish successful output-file persistence. The planned producer test additionally compares the actual output row.

The p739 producer controls are prospective; fixed identity/duplicate coverage and actual Rust elaboration remain unmeasured.

No blocking source defect was identified in this review. The next dependency is the frozen producer qualification; this review allocates no compute.

Final test-only amendment explicitly requires the archived original to be verified with planted scalar recovery before comparing rows. The forced helper probe now retains a separate fresh result and trace. Native driver and adapter bytes are unchanged.

Opt-in diagnostic driver: source-reviewed paths

advance begin → native advance → result → counters/labels

lift begin → native lift → signed disposition → graph result

Trace first failure → incomplete prefix → invocation stops

Proposed: 11 target runs + 2 probes + 4 protocol negatives

Source-only diagram; no candidate runtime measurements.